

Practical-01: --

Code Output: -

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aswitha_patel@Aswitha:~$ nano prac_01.c
aswitha_patel@Aswitha:~$ gcc prac_01.c -o prac_01
aswitha_patel@Aswitha:~$ ./prac_01
Enter a Linux command: ls
Parent process
Parent process id is 795
My parent process id is 596
Child process
Child process id is 796
Parent process id is 795
123.txt  def.c      exececamp3.c  forkexample  openex      pipeex1.c  prac1      readwr.c
a.out    dir3        f1            forkexample.c openex.c    pipeex2    prac1.c    rrt
a2       dir5        f10           linux_task1  permission_lab pipeex2.c  prac_01    sim
a3       dir6        f10.c        linux_task2  pipe10      pr         prac_01.c  sim.c
a6       dirz        f2           linux_task4  pipe10.c    pr.c       practice   student_portal
a7       ex1         f3           mvex         pipe11      pr1        practice.c taskex1
a9       exececamp1 f6           mvex.c       pipe11.c    pr1.c      practice1  taskex1.c
abc      exececamp1.c f6.c        open1        pipe12.c.save pr3        practice1.c xc
b1       exececamp2  f9          open1.c     pipe13      pr3.c     readandopen xyz
b2       exececamp2.c fork4       open3        pipe13.c    prac01     readandopen.c z7
def      exececamp3  fork4.c     open3.c     pipeex1     prac01.c  readwr     z8
Child process completed
aswitha_patel@Aswitha:~$ |
```

B)

Linux terminal commands

uname

```
aswitha_patel@Aswitha:~$ uname
Linux
aswitha_patel@Aswitha:~$ |
```

lscpu

```

aswitha_patel@Aswitha:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 16
On-line CPU(s) list:    0-15
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1340P
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     8
Socket(s):              1
Stepping:               2
BogoMIPS:               4377.61
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
1 xsave avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
ear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full

Caches (sum of all):
L1d:                    384 KiB (8 instances)
L1i:                    256 KiB (8 instances)
L2:                     10 MiB (8 instances)
L3:                     12 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-15
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:        Not affected
Old microcode:          Not affected
Reg file data sampling:  Mitigation; Clear Register File
Retbleed:               Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_D
IS_S
Srbds:                  Not affected
Tsa:                    Not affected
Tsx async abort:        Not affected
Vmscape:                Not affected
aswitha_patel@Aswitha:~$

```

lsblk

```

aswitha_patel@Aswitha:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
aswitha_patel@Aswitha:~$

```

ps

```

aswitha_patel@Aswitha:~$ ps
  PID TTY          TIME CMD
  508 pts/0        00:00:00 bash
  579 pts/0        00:00:00 ps
aswitha_patel@Aswitha:~$

```

top

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24204	15340	11552	S	0.0	0.2	0:00.87	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	42204	16636	15360	S	0.0	0.2	0:00.16	systemd-journal
81	systemd+	20	0	22416	14488	11988	S	0.0	0.2	0:00.10	systemd-resolve
90	root	20	0	35412	12360	9228	S	0.0	0.2	0:00.31	systemd-udev
139	root	20	0	2888	1944	1820	S	0.0	0.0	0:00.08	chronyd-starter
144	root	20	0	4472	3120	2868	S	0.0	0.0	0:00.01	cron
147	message+	20	0	8816	5404	4724	S	0.0	0.1	0:00.06	dbus-daemon
175	root	20	0	44096	29812	14672	S	0.0	0.4	0:00.33	networkd-dispat
186	root	20	0	18420	9388	8156	S	0.0	0.1	0:00.11	systemd-logind
192	root	20	0	1866288	15548	12552	S	0.0	0.2	0:00.14	wsl-pro-service
207	syslog	20	0	220548	5788	4512	S	0.0	0.1	0:00.08	rsyslogd
246	root	20	0	123460	32656	18056	S	0.0	0.4	0:00.27	unattended-upgr
256	_chrony	20	0	20424	6456	5680	S	0.0	0.1	0:00.03	chronyd
262	_chrony	20	0	12096	2404	1760	S	0.0	0.0	0:00.00	chronyd
270	root	20	0	5492	2764	2580	S	0.0	0.0	0:00.00	agetty
355	root	20	0	8648	5528	4684	S	0.0	0.1	0:00.01	login
397	aswitha+	20	0	22336	13492	10880	S	0.0	0.2	0:00.08	systemd
399	aswitha+	20	0	22912	4128	2128	S	0.0	0.1	0:00.00	(sd-pam)
425	aswitha+	20	0	6136	5448	3864	S	0.0	0.1	0:00.02	bash
502	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
503	root	20	0	3200	1128	980	S	0.0	0.0	0:00.00	Relay(508)

aswitha_patel@Aswitha:~\$ top
top - 07:00:06 up 2 min, 1 user, load average: 0.01, 0.02, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7761.6 total, 6488.0 free, 570.7 used, 855.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7190.9 avail Mem

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6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	42204	16636	15360	S	0.0	0.2	0:00.16	systemd-journal
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90	root	20	0	35412	12360	9228	S	0.0	0.2	0:00.31	systemd-udev
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425	aswitha+	20	0	6136	5448	3864	S	0.0	0.1	0:00.02	bash
502	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader